

# World *Robotics* Industrial Robots --- 2026



Statistics, Market Analysis, Forecasts and Case Studies

# World Robotics Industrial Robots 2026

## World Robotics 2026 – Industrial Robots

Produced by VDMA Services GmbH, Lyoner Str. 18, 60528 Frankfurt, Germany.

The robot statistics are based on consolidated world data reported by robot suppliers as well as on the statistics and support of the national robotics associations of North America (A3), Spain (AER), United Kingdom (BARA), People's Republic of China (CRIA), Denmark (DIRA), Japan (JARA), Republic of Korea (KAR), Italy (SIRI), Sweden (SWIRA), and Chinese Taipei (TAIROA).

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We express our most sincere gratitude to all partners!

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## Executive Summary World Robotics 2026 Industrial Robots

Worldwide annual installations of industrial robots:

- **2025: 603,307 units (+11%)**
- **2026: 655,000 units (+9%)**
- **2029: 806,000 units (+7% CAGR from 2026 to 2029)**

### Robot installations 2025: from plateau to new record

**603,307 robots installed in 2025. Annual robot installations leap beyond the 600,000-unit mark.**

After four years of around 550,000 new industrial robot installations, global robot demand accelerated in 2025. The previous record level of 552,946 units achieved in 2022 was exceeded by 9%. **Strong demand from the electronics industry, the automotive industry, and from most parts of the general industry** yielded an 11% year-over-year growth of unit installations in 2025.

Despite substantial regional differences, **the global outlook remains positive**. Installations are forecast to rise by 9% to 655,000 units in 2026 and to reach 806,000 units in 2029, corresponding to average annual growth of 7% from 2026 to 2029. **There is no indication that the overall long-term growth trend will come to an end any time soon.**

The global **operational stock** of industrial robots grew by 9% to **5,079,078 units in 2025**. It is expected to grow by 7% on average each year from 2026 to 2029. After breaking the five-million-unit mark in 2025, the operational stock of industrial robots will exceed six million units in 2028.

In 2025, the **market value** increased by 11% to **US\$18.6bn**.

### Growth in Asia and in the Americas, contraction in Europe

Robot demand in **Asia is booming**: In 2025, **457,315 units** were installed, **up 14%** year-over-year. 76% of all newly deployed robots were installed in Asia, up 2 percentage points from 2024. Three of the top five markets for industrial robots are in Asia: **China is the growth driver and by far the largest market with 354,226 units (+20%) installed in 2025**. The share of domestic suppliers was around 30% from 2019 to 2021 and began to accelerate steeply in 2022. Since 2024, **more than half of the Chinese installations are from Chinese brands**.

The second largest market, **Europe**, experienced a **weak** year with a **14% contraction** of robot demand down to **73,054 units** in 2025. Robot demand in this region also benefited from the nearshoring trend but recently suffered from **weak demand by the automotive industry**.

In **the Americas**, installations were **up by 14%** to a **new record level of 57,044 units**. The **United States** is the largest American market and accounted for 67% (-1 pp) of the installations (38,428 units; +12%) in 2025.

### **Installations and stock of industrial robots from 2020–2025 by customer industries**

For many years, the automotive industry's pattern of purchasing robots largely determined the overall investment pattern in robot installations. This has changed on a global scale and at least in some Asian countries (China, Japan, and the Republic of Korea), where the electrical/electronics industry has become the largest customer industry.

The latest switch took place in 2024, when the **electrical/electronics** industry reclaimed the lead that it maintained in 2025. **Installations were up 10% to 142,235 units**. The **automotive industry** remained the second largest customer of industrial robots in 2025, just around 3,800 units shy of the electronics industry. Demand was **up 10% to a new peak level of 138,429 units**. The **metal and machinery industry** established itself as the third largest customer industry and accounted for 18% (+2 pp) of all installations in 2025. In 2025, installations were **up 22% to a new peak level of 108,571 units**.

### **Regionalization and market access**

Geopolitical tensions, tariffs, non-tariff barriers, industrial policies, and supply chain risks are encouraging manufacturers to diversify production and **establish capacity closer to their main markets**. This does not necessarily increase global robot demand, but it changes where investment takes place. For robot suppliers, regionalization increases the importance of local production, engineering, system integration, and service and support infrastructure. Where **protectionist policies** are in place, market access will increasingly depend on local supply chains, local presence, and compliance with national requirements.

Standardized and modular cells, preconfigured application packages, simpler programming, and faster commissioning can **reduce integration costs** and project risks.

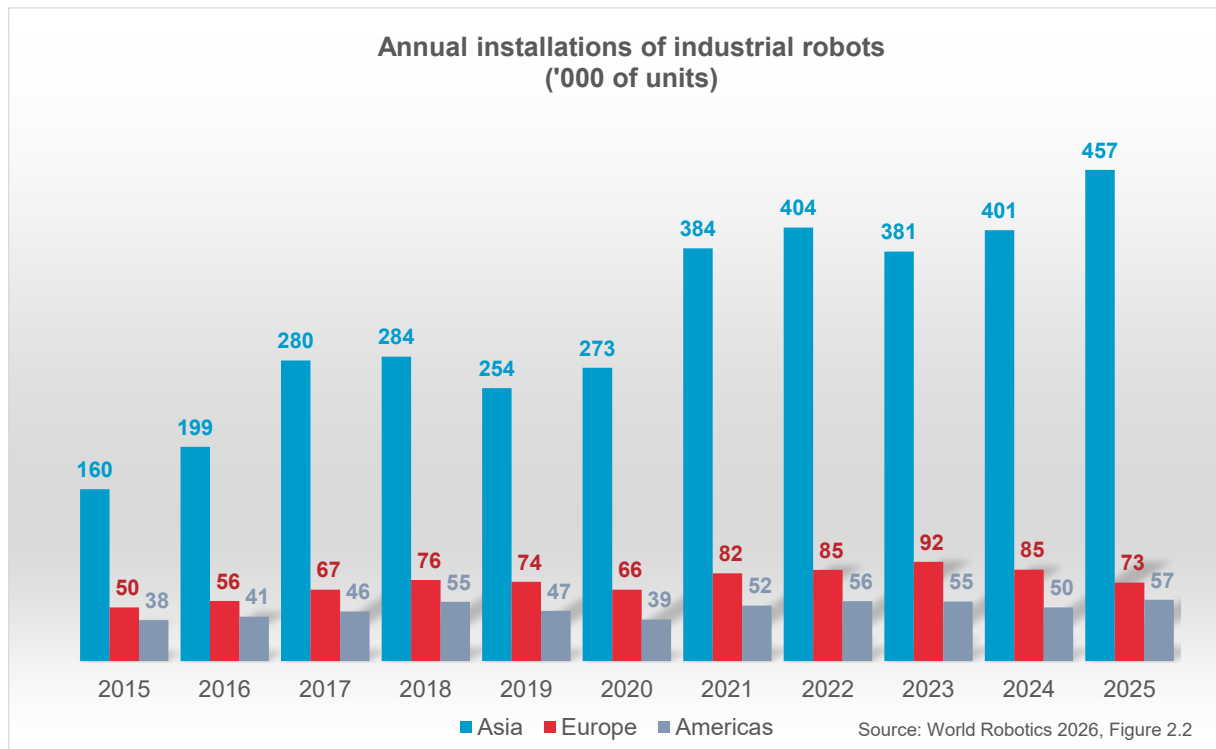
### **Trends in robot technology**

**Artificial intelligence** is becoming a fundamental **enabling technology** across robotics. Improvements in computing power, data availability, and AI models are expanding both the range and the sophistication of its applications.

The **degree of maturity differs considerably by application**. Machine vision for detection and robot guidance is already established and industry-grade. Predictive maintenance, process optimization and AI-based inspection are also increasingly used

in industry. General-purpose Physical AI systems and learning from simulation and experience, are not yet industry-grade.

Most humanoids remain prototypes or pilot-stage technologies rather than industrial-grade products. At the current stage of technology, industrial robots are economically superior for industrial production tasks. Humanoids may eventually complement these systems in selected applications, but widespread industrial deployment will depend on substantial improvements in reliability, safety, operating time, and economic viability.



# 1 Introduction

**Chapter 1 reviews definitions and classifications of industrial robots.**

## 1.2 DATA SOURCES, REVISIONS, RELIABILITY, AND VALIDITY

### 1.2.1 SOURCES

World Robotics Industrial Robots statistics rely on primary and secondary data.

The **primary source** is data on robot installations by country, industry, and application that nearly all major industrial robot suppliers worldwide report to the IFR Statistical Department directly.

Several national robotics associations collect data on their national robot markets and provide their results as **secondary data** to the IFR. These data are used to validate the IFR primary data, thus ensuring data quality. They are also used to fill in the missing information of companies not reporting to the IFR directly. The final statistics provided in this publication and in the online database are therefore the **consolidated primary and secondary data**.

The submission of statistical data on industrial robots is mandatory for robot suppliers (IFR Constitution § C3.4) and national robot associations (IFR Constitution § C3.3) in the IFR. All other robot suppliers are invited to participate directly as a primary source or indirectly through national associations.

Since 2015, the Korean Association of Robot Industry (KAR) has been providing data on Korean production, imports, and exports. Before 2015, the Korean Machine Tool Manufacturers Association (KOMMA) provided the data on the Korean robot companies' sales in the Republic of Korea.

The Japanese Robot Association (JARA) provides consolidated global data for Japanese companies' robot shipments in compliance with the IFR industry and application classifications. They also provide Japanese production, imports and exports. Until 2017, JARA contributed national statistics for Japan including exports. The information was refined using data that the IFR Statistical Department received directly from European subsidiaries of Japanese companies and for North America by results from A3 shipment statistics. JARA also reports the operational stock of robots in Japan. When working with historic data (access is included in the Premium version of World Robotics), note that there is a break in the time series for data on Japan between 2000 and 2001 resulting from international harmonization of definitions and coverage of statistics.

The Association for Advancing Automation (A3, formerly: Robotic Industries Association, RIA) provides data on North America. The statistics report shipment data from North American companies in compliance with the IFR industry and application classifications. Primary data, not included in the A3 statistics and JARA data on exports of Japanese suppliers to North America, supplement these data. Until and including 2010, data were available only for North America (Canada, Mexico, United States) as a whole. From 2004 to 2010, data on North America were based on the consolidated data of robot suppliers

worldwide and JARA. Before 2004, the annual data for North America only comprised what was reported to RIA by its member companies.

Since 2013, the Chinese Robot Industry Alliance (CRIA) provides installations of Chinese robot suppliers on the domestic market in compliance with the IFR industry and application classifications. Since the reporting year 2018, CRIA also reports export data of Chinese suppliers, but only by country and type. Due to legal requirements, CRIA can only report member statistics.

IFR also gratefully appreciates the support of the national robotics associations of Spain (AER), Italy (SIRI), Sweden (SWIRA), Denmark (DIRA) and Chinese Taipei (TAIROA).

Prior 2004, country reports relied exclusively on data of national robot associations. This holds true for Denmark, Finland, France, Germany, Italy, Japan, Rep. of Korea, North America, Norway, and Spain. Reports on other countries were based on data provided by only a few robot suppliers. In 2005, robot suppliers reported consolidated data classified by country, industry, or application for the first time. This facilitated more detailed reports on countries that do not have a national robotics association.

For sources of **employment data** and methods of computing robot densities, see chapter 2.5.1.

### 1.2.2 QUALITY, RELIABILITY, AND VALIDITY

IFR Statistical Department considers the high-quality data to be valid and reliable for the following reasons:

IFR Statistical Department provides definitions and delimitations of robot types, industries and applications to all primary and secondary data sources to ensure data reliability.

The objective of World Robotics Industrial Robots is a comprehensive overview of the dissemination of industrial robots globally. The main indicator is the number of robots newly installed per year. The implications of some companies and associations counting “shipments” instead of “installations” are discussed in chapter 1.5. Some associations also provide information on sales in monetary units for their territory. IFR Statistical Department considers these indicators a valid measure of robot dissemination.

The statistics presented in World Robotics Industrial Robots cover almost the whole population. This is ensured by permanent market observation and cooperation with national robotics associations. The availability of primary and secondary data sources enables IFR Statistical Department to check the data for consistency. This makes the data reliable.

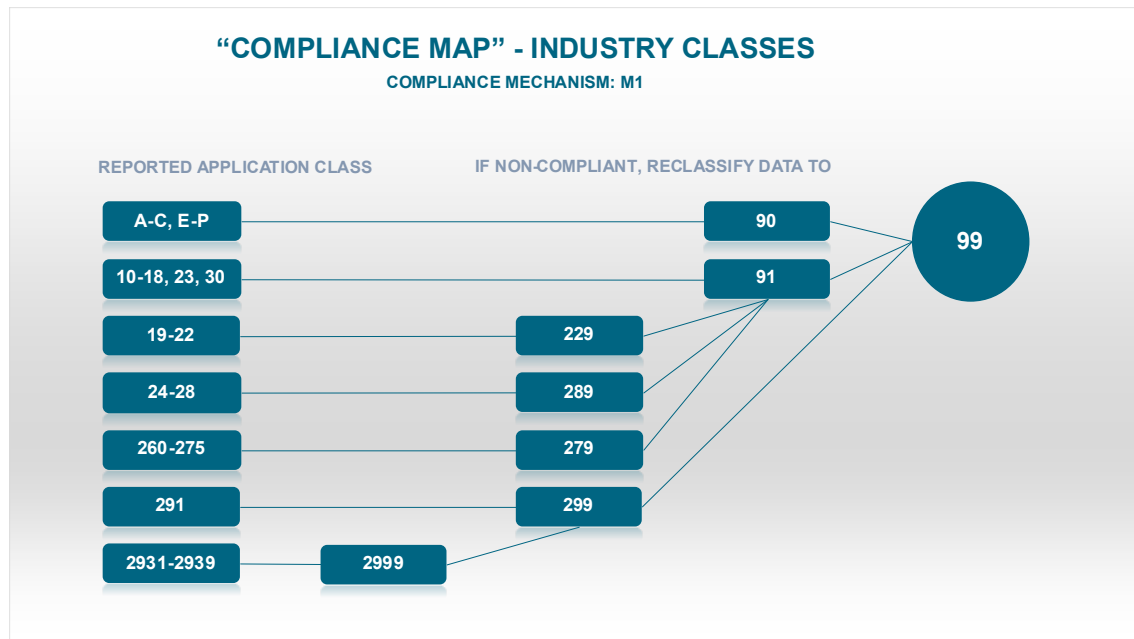


Figure 1.3: Compliance re-classification of data by industry class.

#### 1.4 FORECASTS

Forecasts of investments in industrial robots, presented in chapter 4, are derived as the consolidated assessment of:

- economic factors
- technological progress
- expert opinions of some of the leading robot manufacturers, major robot users and national robotics associations

### 1.5 DEFINITION OF “SHIPMENTS” AND “INSTALLATIONS”

The IFR surveys annual installations of industrial robots. However, as a compromise between the theoretical idea of counting the actual installation of the robot at the customer's site and the practical data availability which often refers to the shipment of the robot, reporting of shipment data is also accepted. Note that shipment and installation data might differ for several reasons: A geographical measurement bias might occur for instance, if a reseller or integrator exports a robot and does not share this information with the robot producer. The time of installation might be subject to measurement bias if a robot is shipped, but not installed in the same year, e.g. because it is still en route or went to the inventory or is work-in-progress at the integrator's site.

## 1.6 DEFINITION OF “OPERATIONAL STOCK”

The operational stock of robots measures the number of robots currently deployed. JARA calculates and provides this number for Japan. For other countries, IFR Statistical Department calculates the operational stock assuming an average service life of 12 years with an immediate withdrawal from service afterwards. This assumption was investigated in an UNECE/IFR pilot study, carried out in 2000 among some major robot companies (see annex B in World Robotics 2001).<sup>1</sup> This investigation suggested that an assumption of 12 years of average life span might be too conservative and that the average service life was closer to 15 years. On the other hand, German tax authorities suggest in their standard depreciation schedules an average service life of 5 years for robots in the automotive industry and 6 years for robots in the mechanical engineering industry. Similarly, useful life of class 80.0C “Robotics” is 5 years in the American tax law. Of course, robots may be refurbished and appreciated, so the standard depreciation schedule underestimates the service life. Nevertheless, the differences (5, 12 or 15 years) are substantial and need further investigation. Presumably, there are substantial differences depending on industry, application, and type of robot. In the meantime, the operational stock is calculated as the sum of robot installations over 12 years.

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<sup>1</sup> For several years IFR and the United Nations Economic Commission for Europe UNECE have cooperated closely in the compilation, processing and analysis of worldwide statistics on industrial robots. In 2005, the full responsibility for World Robotics was transferred to the IFR Statistical Department.

## 1.7 DEFINITIONS: “ROBOT”, “SERVICE ROBOT”, “INDUSTRIAL ROBOT”, “MEDICAL ROBOT”

### 1.7.1 ISO 8373:2012 VOCABULARY DEFINITIONS

ISO 8373:2021 “*defines terms used in relation to robotics*” (§ 1). These vocabulary definitions relate to industrial robotics, service robotics, and medical robotics. This section describes the ISO definitions needed to understand IFR classifications schemes, IFR industrial robot statistics, and the delimitation to IFR service robot statistics.

According to ISO 8373:2021, a **robot** is a “*programmed *actuated mechanism with a degree of autonomy, to perform locomotion, manipulation or positioning*” (§ 3.1). Mechanisms that are using robot technology but do not satisfy the definition of robots, for instance teleoperated manipulators, are called **robotic device**. (§ 3.5). Robot technology includes perception, reasoning and planning algorithms (§ 3.3).*

A service robot is a robot “that performs useful tasks for humans or equipment” (§ 3.7).

A medical robot is a “robot intended to be used as medical electrical equipment” (§ 3.8).

An **industrial robot** is an “*automatically controlled, reprogrammable multipurpose manipulator, programmable in three or more axes, which can be either fixed in place or fixed to a mobile platform for use in automation applications in an industrial environment*” (§ 3.6).

That is, industrial robots have **at least three axes** (directions used to specify the robot motion in a linear or rotary mode, § 5.3) and they satisfy all of the following:

- **Automatically controlled:** The robot control system operates in accordance with the task program (§ 6.13.2). The task program is a set of instructions for motion and auxiliary functions that define the specific intended task (§ 6.1). The opposite of automatically controlled operation is manual operation, where a human operator uses input devices (e.g. joysticks or pushbuttons) to control the motions of the machine (§ 6.13.1).
- **Reprogrammable:** designed so that the programmed motions or auxiliary functions can be changed **without physical alteration**. Physical alterations are alterations of the mechanical system. Note: This interpretation of the word “reprogrammable” was taken from ISO 8373:2012, § 2.3. ISO 8373:2021 does not provide a definition anymore.
- **Multipurpose:** capable of being adapted to a different application with physical alteration. Note: This interpretation of the word “multipurpose” was taken from ISO 8373:2012, § 2.5. ISO 8373:2021 does not provide a definition anymore.
- **Manipulator** (§ 4.14): An arrangement of segments, jointed or sliding relative to one another. The end effector is not part of the manipulator (§ 4.14, Note 2).
- **Fixed in place or to a mobile platform:** The robot can be mounted to a stationary point, but it can also be mounted to a non-stationary point, e.g.

## 1.8 COPE OF IFR INDUSTRIAL ROBOT STATISTICS

IFR industrial robot statistics count multipurpose industrial robots as defined in the previous chapter, only. **Robotic devices** are excluded.<sup>2</sup> IFR industrial robot statistics generally exclude **dedicated industrial robots**. Dedicated industrial robots are industrial robots specifically designed for and controlled by a special machine. However, industrial robots with their own control system, not controlled by the machine, are included, even if they are dedicated to a special machine.

Examples of dedicated industrial robots excluded from the statistics are:

- Equipment dedicated for loading/unloading of machine tools
- Dedicated assembling equipment, e.g. for assembling on printed circuit boards
- Integrated Circuit Handlers (pick and place)
- Automated storage and retrieval systems

**Wafer handlers** have their own control system and are **included** in the industrial robot statistics. Wafers handlers can be articulated, cartesian, cylindrical or SCARA robots. They are reported in the IFR application class 902 “cleanroom for semiconductors”.

**Flat panel handlers are included**, too. These are mainly articulated robots, and they are reported in the application class 901 “cleanroom for FPD”.

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<sup>2</sup> This is in contrast to IFR service robot statistics, which include robotic devices under specific conditions.

### 1.1 IFR CLASSIFICATION OF INDUSTRIES

Starting with World Robotics 2010, data by customer industry has been reported using the IFR industry classification scheme. IFR industry classes are related to and derived from the International Standard Industrial Classification of All Economic Activities (ISIC) revision 4 scheme. It does, however, not exactly correspond to ISIC revision 4. Classes with minor robot installation counts were aggregated and classes that include major customer industries (automotive in particular) were further divided to provide more detail (see table 1.1). Prior to 2010, data was presented according to ISIC revision 2 or 3. All earlier data was transferred to the new classification scheme.

Table 1.1

| IFR class | Categories, divisions and classes of economic activities, ISIC, rev. 4      | Definitions                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A-B       | Agriculture, hunting and forestry; fishing                                  | Crop and animal production, hunting and related service activities, forestry and logging, fishing and aquaculture                                                                                                                                                                                                                                                                                |
| C         | Mining and quarrying                                                        | Mining of coal and lignite, extraction of crude petroleum and natural gas, mining of metal ores, mining support service                                                                                                                                                                                                                                                                          |
| D         | Manufacturing                                                               |                                                                                                                                                                                                                                                                                                                                                                                                  |
| 10-12     | Food products and beverages; Tobacco products                               |                                                                                                                                                                                                                                                                                                                                                                                                  |
| 13-15     | Textiles, leather, wearing apparel                                          | Textiles; Wearing apparel; dressing & dyeing of fur; Luggage, handbags, saddlery, harness and footwear                                                                                                                                                                                                                                                                                           |
| 16        | Wood and wood products (incl. furniture)                                    | Manufacture of wood, products of wood (incl. wood furniture) and products of cork                                                                                                                                                                                                                                                                                                                |
| 17-18     | Paper and paper products, publishing & printing                             | Manufacture of pulp, paper and converted paper production, printing of products, such as newspapers, books, periodicals, business forms, greeting cards, and other materials, and associated support activities, such as bookbinding, plate-making services, and data imaging; reproduction of recorded media, such as compact discs, video recordings, software on discs or tapes, records etc. |
| 19-22     | Plastic and chemical products                                               |                                                                                                                                                                                                                                                                                                                                                                                                  |
| 19        | Chemical products, pharmaceuticals, cosmetics                               | Manufacture of basic pharmaceutical products and pharmaceutical preparations. This includes also the manufacture of medicinal chemical and botanical products                                                                                                                                                                                                                                    |
| 20-21     | Unspecified chemical, petroleum products                                    | Transformation of crude petroleum and coal into usable products, transformation of organic and inorganic raw materials by a chemical process and the formation of products                                                                                                                                                                                                                       |
| 22        | Rubber and plastic products without automotive parts*                       | (e.g. rubber tires, plastic plates, foils, pipes, bags, boxes, doors, etc.) rubber and plastic parts for motor vehicles should be reported in 29.3.2                                                                                                                                                                                                                                             |
| 23        | Glass, ceramics, stone, mineral products n.e.c. (without automotive parts*) | Manufacture of intermediate and final products from mined or quarried non-metallic minerals, such as sand, gravel, stone or clay, manufacture of glass, flat glass ceramic and glass products, clinkers, plasters, etc.                                                                                                                                                                          |
| 24-28     | Metal                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                  |
| 24        | Basic metals (iron, steel, aluminium, copper, chrome)                       | e.g. iron, steel, aluminium, copper, chrome etc.                                                                                                                                                                                                                                                                                                                                                 |
| 25        | Metal products (without automotive parts*), except machinery and equipment  | e.g. metal furniture, tanks, metal doors, forging, pressing, stamping and roll forming of metal, nails, pins, hand tools, etc.                                                                                                                                                                                                                                                                   |
| 28        | Industrial machinery                                                        | e.g. machinery for food processing and packaging, machine tools, industrial equipment, rubber and plastic machinery, industrial cleaning machines, agricultural and forestry machinery, construction machinery etc.                                                                                                                                                                              |

Sources: IFR